# Data-Driven Identification and MPC of Coupled Tank System using Bilinear Koopman Realizations and Physics-Informed Neural Networks

Amir Vanegas*[a], Duvan Tellez-Castro[a], Nelson Díaz-Aldana[a], Julio Barón-Velandia[a]

[a]Universidad Distrital Francisco José de Caldas

*Corresponding/Presenting Author Email: azvanegasc@udistrital.edu.co

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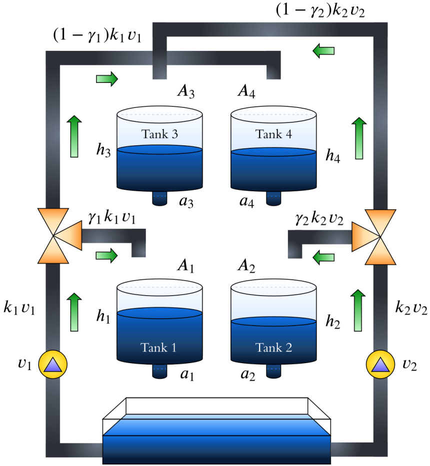
# Outline

# Problem

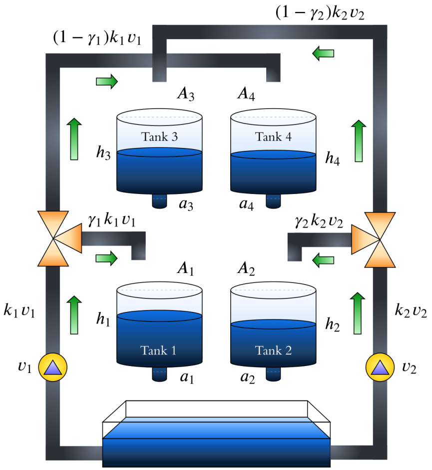



## System Model

- Governed by mass balance differential equations incorporating Torricelli's discharge law

$$\dot h_1 = -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1}\sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1}v_1,$$
$$\dot h_2 = -\frac{a_2}{A_2}\sqrt{2gh_2} + \frac{a_4}{A_2}\sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2}v_2,$$
$$\dot h_3 = -\frac{a_3}{A_3}\sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3}v_2,$$
$$\dot h_4 = -\frac{a_4}{A_4}\sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4}v_1.$$

- Interacts four interconnected water tanks through two independent cross-coupled pumps.

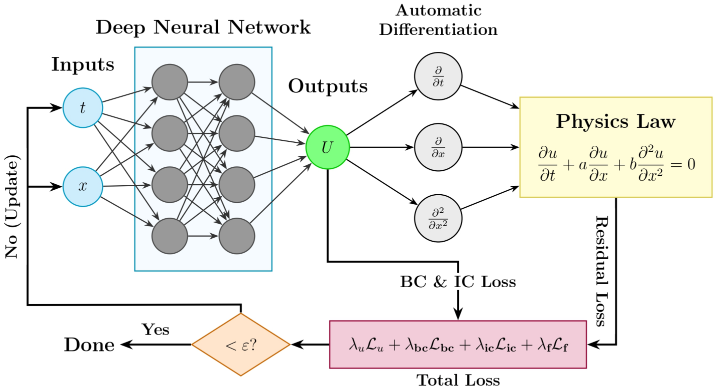



## Control Objective

- Regulate and track desired water levels in the lower tanks $(h_1, h_2)$.

## Control Challenges

- Requires high-fidelity models capturing complex behaviors as a reliable reference for control.
- Presents strong multivariable interactions that severely limit standard decentralized architectures.
- Operates on initial condition $\mathbf{h}_0 = [12.4, 12.7, 1.8, 1.4]^\top$ cm.

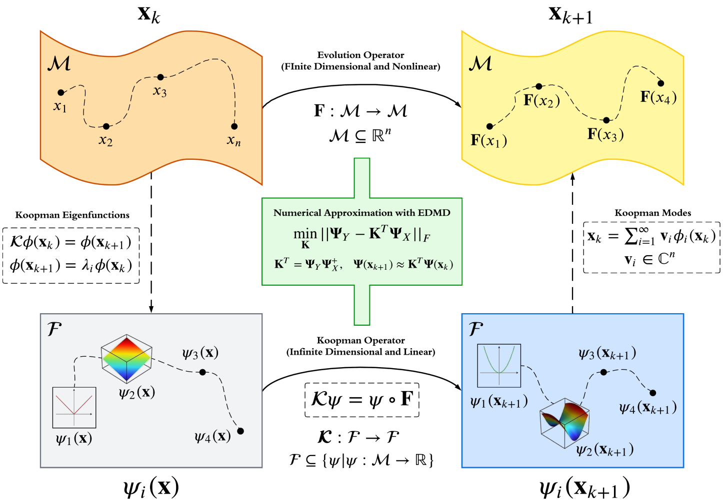



## Overview

- Approximates non-linear solutions as a functional mapping of input variables

$$u = \mathcal{N}(x, t; \theta).$$

- Learns system dynamics by minimizing residuals derived from the governing physical equations.
- Discrete-time variants can be formulated through state-space model representations.




## Overview

- Consider a discrete-time dynamical system defined by

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k),$$

  the Koopman operator $\mathcal{K}$ evolves observables $\psi$ along trajectories via composition.

- It provides a global linear infinite-dimensional representation equivalent to the underlying non-linear dynamics.

# Bilinear Koopman Framework

## Bilinear Lifting Realization

- Maps the physical space into a higher-dimensional state $\mathbf{z}_k = \mathbf{\Psi}(\mathbf{h}_k)$, where the state evolves according to the bilinear dynamics

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \sum_{j=1}^{m} (\mathbf{B}_j \mathbf{z}_k) u_{j,k} + \mathbf{D}\mathbf{u}_k.$$

- Linear projection back to physical tank levels through the output matrix $\mathbf{C}$ in

$$\hat{\mathbf{h}}_k = \mathbf{C}\mathbf{z}_k.$$

- Dictionary of observables used to approximate the Koopman operator

$$\mathbf{\Psi}(\mathbf{h}) = [\mathbf{h}, \sqrt{|\mathbf{h}|}, \sin(\mathbf{h}), \cos(\mathbf{h}), \mathbf{h}^2, |\mathbf{h}|, \phi_{\mathrm{RBF}}(\mathbf{h})]^\top.$$

## Bilinear Identification with EDMD

- **Step 1:** Solves the evolution weights $\mathbf{W}_{dyn} = [\mathbf{A}, \mathbf{B}_1, \ldots, \mathbf{B}_m, \mathbf{D}]$ via Ridge regression over the data matrix $\mathbf{X}_{reg}$

$$\min_{\mathbf{W}_{dyn}} \|\mathbf{Z}_{next} - \mathbf{W}_{dyn}\mathbf{X}_{reg}\|_F^2 + \alpha_1 \|\mathbf{W}_{dyn}\|_F^2.$$

- **Step 2:** Identifies $\mathbf{C}$ through a separate regularized least-squares optimization.

$$\min_{C} \|\mathbf{H} - \mathbf{C}\mathbf{Z}\|_F^2 + \alpha_2 \|\mathbf{C}\|_F^2.$$

## Radial Basis Functions

$$\phi_j(\mathbf{h}) = \exp(-\|\mathbf{h} - \mathbf{c}_j\|^2/\sigma).$$

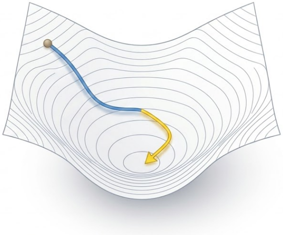
# PINN Identification Framework

## Properties

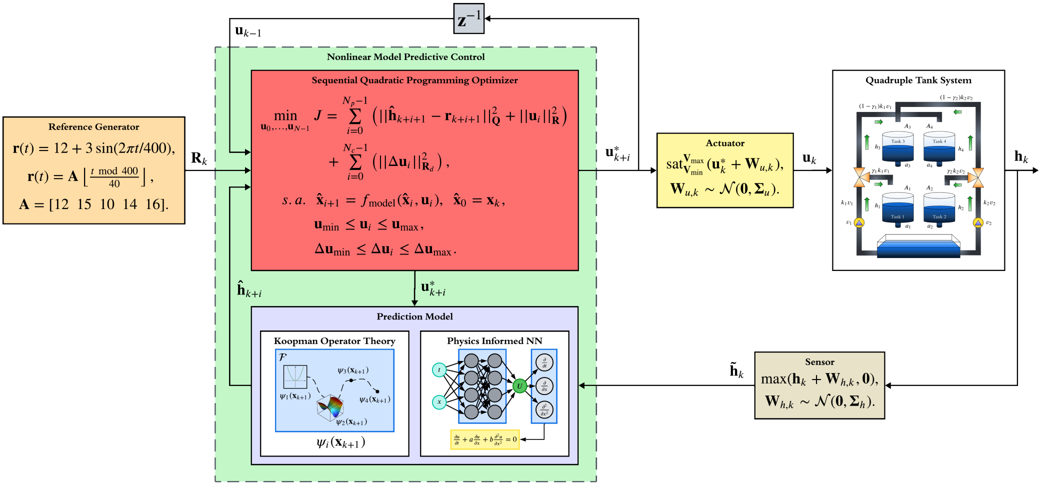
- Minimizes a composite loss function balancing data accuracy, physics consistency, and initial conditions
$$\mathcal{L}(\boldsymbol{\theta}) = w_d\mathcal{L}_{data} + w_p\mathcal{L}_{phys} + w_{ic}\mathcal{L}_{ic}.$$
- Penalizes trajectories deviating from the discrete-time mass balance over the hydraulic manifold
$$\mathbf{R}_k = \hat{\mathbf{x}}_{k+1} - \left(\mathbf{x}_k + \Delta t \cdot \mathbf{f}_{phys}(\mathbf{x}_k, \mathbf{u}_k)\right).$$
- Executes data normalization and denormalization pipelines to ensure stable gradient propagation.

## Hybrid Optimization Pipeline

- **Adam:** Escapes local minima and navigates rugged loss landscapes efficiently during early training phases.
- **L-BFGS-B:** Computes exact descent steps to exploit smooth physical boundaries and achieve tight residual convergence.



# Nonlinear Model Predictive Control (NMPC)



# Quadruple Tank Identification

## Data Acquisition

- Simulates hydraulic dynamics for 5000 s at $dt = 0.05$ s via RK45 integration.
- Generates persistent excitation using PRBS voltage signals switching every 50 s.
- Injects white Gaussian noise $\sigma = 0.1$ to emulate plant operating conditions.

## Data Preprocessing

- Leverages a Kalman filter to mitigate noise prior to identification.
- Employs the Optuna framework for hyperparameter optimization.

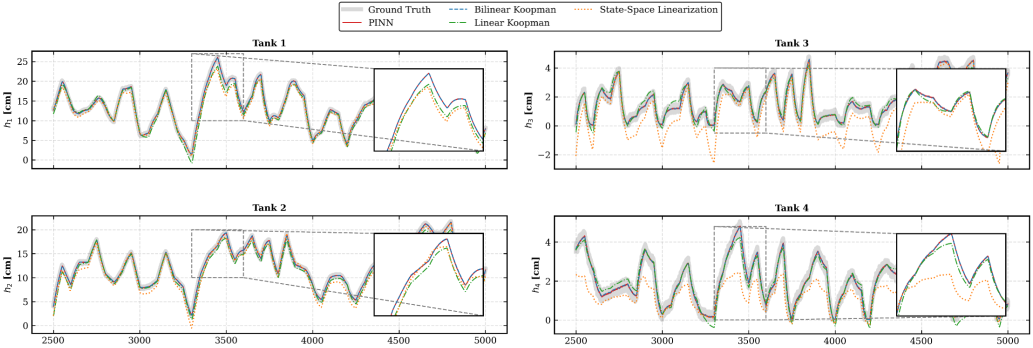
PHYSICAL PARAMETERS OF THE QUADRUPLE TANK SYSTEM.

| **Symbol** | **Description** | **Value** | **Unit** |
|---|---|---|---|
| $A_{1,3}$ | Area (Tanks 1 & 3) | 28.00 | $cm^2$ |
| $A_{2,4}$ | Area (Tanks 2 & 4) | 32.00 | $cm^2$ |
| $a_{1,3}$ | Outlet area (Tanks 1 & 3) | 0.071 | $cm^2$ |
| $a_{2,4}$ | Outlet area (Tanks 2 & 4) | 0.057 | $cm^2$ |
| $k_1, k_2$ | Pump flow constants | 3.33, 3.35 | $cm^3$/Vs |
| $\gamma_1, \gamma_2$ | Flow split ratios | 0.70, 0.60 | – |
| $g$ | Gravity | 981.00 | $cm/s^2$ |

PARAMETERS FOR KOOPMAN AND PINN IDENTIFICATION

| **Koopman Parameter** | **Value** | **PINN Parameter** | **Value** |
|---|---|---|---|
| Observables ($n_{obs}$) | 32 | Hidden Layers | 4 |
| Regularization ($\alpha_i$) | $8.8 \times 10^{-7}$ | Neurons per Layer | 48 |
| RBF bandwidth ($\sigma$) | 20 | Activation Function | SiLU |
| RBF centers count ($M$) | 8 | Adam Epochs | 4000 |
| $\mathbf{K}$ matrix size | $4 \times 32$ | L-BFGS Max Iter. | 4000 |

# Identification Results




| Model | Metric | $h_1$ | $h_2$ | $h_3$ | $h_4$ |
|---|---|---|---|---|---|
| PINN | RMSE (cm) | **0.2061** | 0.1770 | 0.1228 | 0.1125 |
| | VAF (%) | **99.85** | **99.83** | 99.06 | 99.03 |
| Bilinear Koopman | RMSE (cm) | 0.2107 | **0.1764** | **0.1222** | **0.1114** |
| | VAF (%) | 99.84 | **99.83** | **99.07** | **99.05** |

| Model | Metric | $h_1$ | $h_2$ | $h_3$ | $h_4$ |
|---|---|---|---|---|---|
| Linear Koopman | RMSE (cm) | 0.8986 | 0.7192 | 0.3322 | 0.2543 |
| | VAF (%) | 98.35 | 98.75 | 93.25 | 95.28 |
| State-space Linearization | RMSE (cm) | 1.3188 | 1.2000 | 1.1826 | 0.8995 |
| | VAF (%) | 96.24 | 96.59 | 59.26 | 75.24 |

# Quadruple Tank Predictive Control

## Validation Scenarios (1500 s)

- Executes 10 closed-loop simulations per model varying the seed for statistical consistency.

## Real-Time Execution

- Operates in closed-loop at $\Delta t = 1\,\mathrm{s}$ using CasADi.
- Predicts system behavior up to 25 s ahead to calculate optimal actions.
- Evaluates robustness via Gaussian noise injected into sensors $\sigma_s = 0.1$ and actuators $\sigma_a = 0.15$.

NMPC Hyperparameters

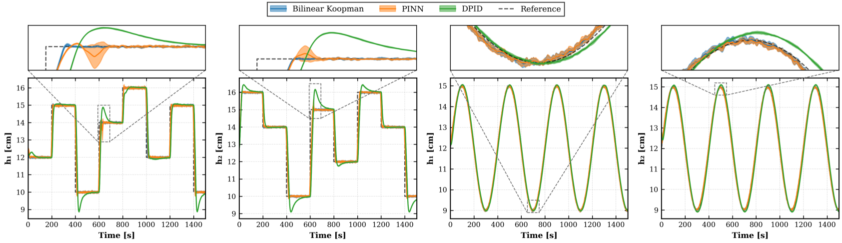
| Parameter | Value | Description |
|---|---|---|
| $N_p$ | 25 | Prediction horizon [steps] |
| $N_c$ | 1 | Control horizon [steps] |
| $\Delta\mathbf{u}_{\max}$ | 0.5 | Slew-rate limit [V] |
| *Weighting Matrices* | | |
| $q_{1,2}$ | 500 | Tracking weights $(h_1, h_2)$ |
| $q_{3,4}$ | 0 | Stability weights $(h_3, h_4)$ |
| $\mathbf{R}$ | $0.01 \cdot \mathbf{I}$ | Control effort weight |
| $\mathbf{R}_d$ | $1.0 \cdot \mathbf{I}$ | Slew-rate weight |

## Control Scenarios

$$\mathbf{r}(t) = 12 + 3\sin(2\pi t/400),$$

$$\mathbf{r}(t) = [12 \quad 15 \quad 10 \quad 14 \quad 16] \cdot \left\lfloor \frac{t \bmod 400}{40} \right\rfloor .$$

# Control Performance Results




GLOBAL PERFORMANCE METRICS FOR MPC COMPARISON.

| Scenario | Metric | Unit | Bilinear Koopman | | | PINN | | | DPID | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Mean | Std | 99% CI | Mean | Std | 99% CI | Mean | Std | 99% CI |
| Sinusoidal Tracking | RMSE | cm | **0.0955** | 0.0012 | (0.0943, 0.0968) | **0.0955** | 0.0011 | (0.0944, 0.0966) | 0.1207 | 0.0006 | (0.1201, 0.1213) |
| | NRMSE | - | **0.0159** | 0.0002 | (0.0157, 0.0161) | **0.0159** | 0.0002 | (0.0157, 0.0161) | 0.0201 | 0.0001 | (0.0200, 0.0202) |
| | ISE | $cm^2s$ | **27.382** | 0.7067 | (26.656, 28.109) | 27.444 | **0.5943** | (26.833, 28.054) | 47.787 | 0.4529 | (47.321, 48.252) |
| | RTF | - | **22.38** | 0.6957 | (21.67, 23.10) | 10.68 | 0.2861 | (10.39, 10.98) | 9.51 | 0.2093 | (9.29, 9.72) |
| Step Response Cascade | RMSE | cm | **0.4403** | 0.0040 | (0.4362, 0.4445) | 0.4528 | 0.0061 | (0.4465, 0.4590) | 0.8224 | 0.0031 | (0.8192, 0.8256) |
| | NRMSE | - | **0.0734** | 0.0007 | (0.0727, 0.0741) | 0.0755 | 0.0010 | (0.0744, 0.0765) | 0.1371 | 0.0005 | (0.1365, 0.1376) |
| | ISE | $cm^2s$ | **595.84** | 10.761 | (584.79, 606.90) | 624.28 | 15.697 | (608.15, 640.41) | 2032.3 | 15.161 | (2016.7, 2047.9) |
| | RTF | - | **23.24** | 1.1716 | (22.03, 24.44) | 10.78 | 0.0633 | (10.71, 10.84) | 10.24 | 0.3143 | (9.91, 10.56) |

*Note:* Calculations derive the 99% confidence intervals (CI) using a Student's $t$-distribution based on a sample size of $n = 10$ stochastic simulations.

# Conclusions

- **High Fidelity:** Both data-driven paradigms capture nonlinear dynamics successfully, achieving mean VAF values above 99%.
- **Noise Resilience:** Bilinear Koopman effectively filters high-frequency disturbances, outperforming the noise-sensitive PINN architecture.
- **Real-Time Velocity:** Koopman NMPC doubles PINN speed during step transients with a Real-Time Factor above 22.
- **Structural Limits:** Scaling exposes neural exploding gradients in PINNs and operator instability risks in Koopman lifting.

**Future Work:** Explore sparse identification and robust stabilization constraints under broader operational regimes.

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# Thank You


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CoDiT_Comparison-of-PINN-and-Bilinear-Koopman